

Retail Customer Behavior & Churn Prediction Report

1. Objective

Analyze customer purchasing behavior and predict churn using RFM analysis and machine learning.

2. Data Source

Online Retail Dataset containing transactional data including Invoice, Quantity, Price, CustomerID, and Date.

3. Data Cleaning

Removed missing CustomerID, negative Quantity, and invalid UnitPrice. Created TotalPrice feature.

4. SQL Analysis

Performed revenue analysis, customer segmentation, top customers, and monthly trends using PostgreSQL.

5. Feature Engineering

Generated RFM features: Recency, Frequency, Monetary. Created AOV (Average Order Value).

6. Churn Definition

Used data-driven threshold based on recency distribution to classify churn.

7. Model Building

Built Logistic Regression and Random Forest models. Evaluated using ROC-AUC and classification metrics.

8. Key Insights

- Revenue is concentrated among a small percentage of customers.
- Majority of customers are low-frequency buyers.
- High recency customers show higher churn risk.
- Frequency and monetary value indicate customer loyalty.
- Model enables risk-based customer segmentation.

9. Conclusion

RFM features effectively capture customer behavior. The model predicts churn probability and enables proactive retention strategies. Improving retention of high-value customers can significantly increase revenue.